An empirical example: Four terminals and two areas

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2025 / 01 / 05

This is a worked example using a four terminals tree, distributed in two areas. Initially, all branches are equal.

```r
# Four taxa and two areas


## Load libraries

library(blepd)
```

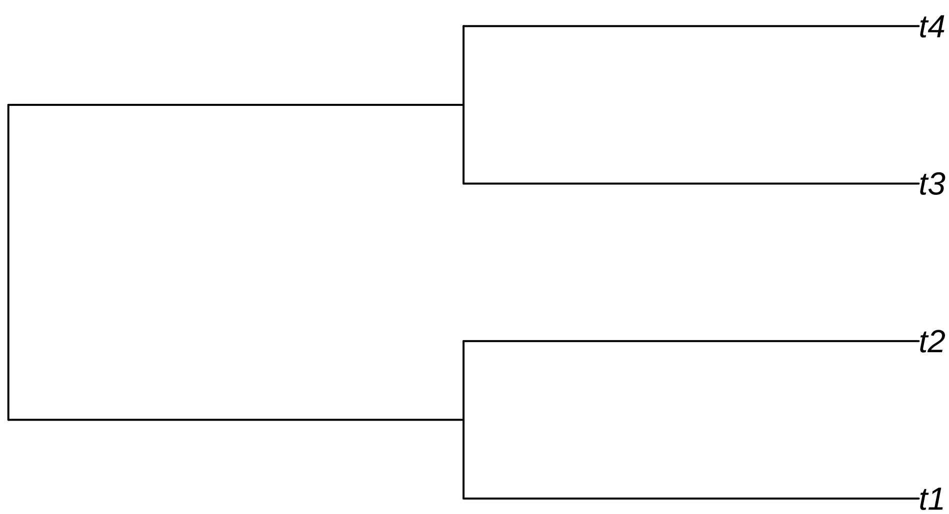
```
## Loading required package: ape

## Loading required package: picante

## Loading required package: vegan

## Loading required package: permute

##
## Attaching package: 'permute'

## The following object is masked from 'package:devtools':
##
##     check

## Loading required package: lattice

## This is vegan 2.6-8

## Loading required package: nlme
```

```r
library(ggtree)
```

```
## ggtree v3.14.0 Learn more at https://yulab-smu.top/contribution-tree-data/
##
## Please cite:
##
## Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
## ggtree: an R package for visualization and annotation of phylogenetic
## trees with their covariates and other associated data. Methods in
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628

##
## Attaching package: 'ggtree'

## The following object is masked from 'package:nlme':
##
##     collapse
```

```
## The following object is masked from 'package:ape':
##
##     rotate
```

```
library(ggplot2)



## Call the data

# Call the four terminals example

##data(package = "blepd")

## trees

data(tree)


## Plot the tree

plot(tree,use.edge.length =T)
```



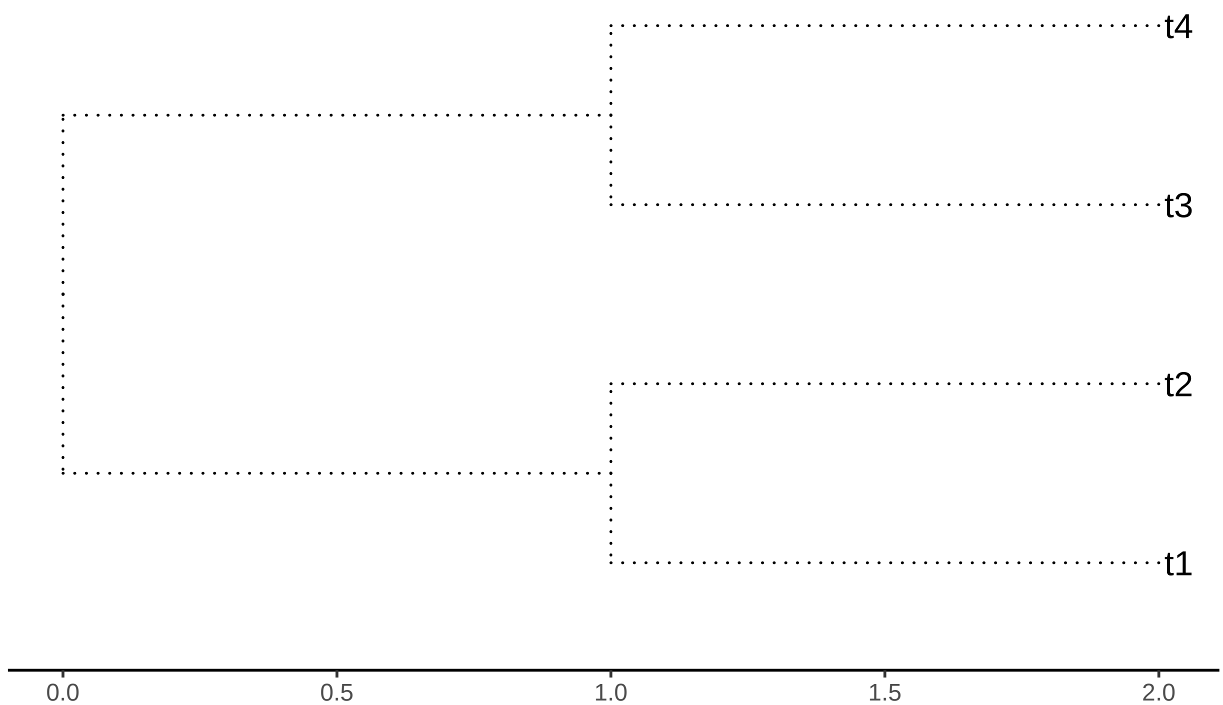

```
initialTree <- tree



## Load and plot the distribution
```

```r
data(distribution)

##str(distribution)

dist4taxa <- distribution

## Change distribution to XY to be able to plot the data, otherwise, skip the next step

distXY <- matrix2XY(dist4taxa)

distXY
```

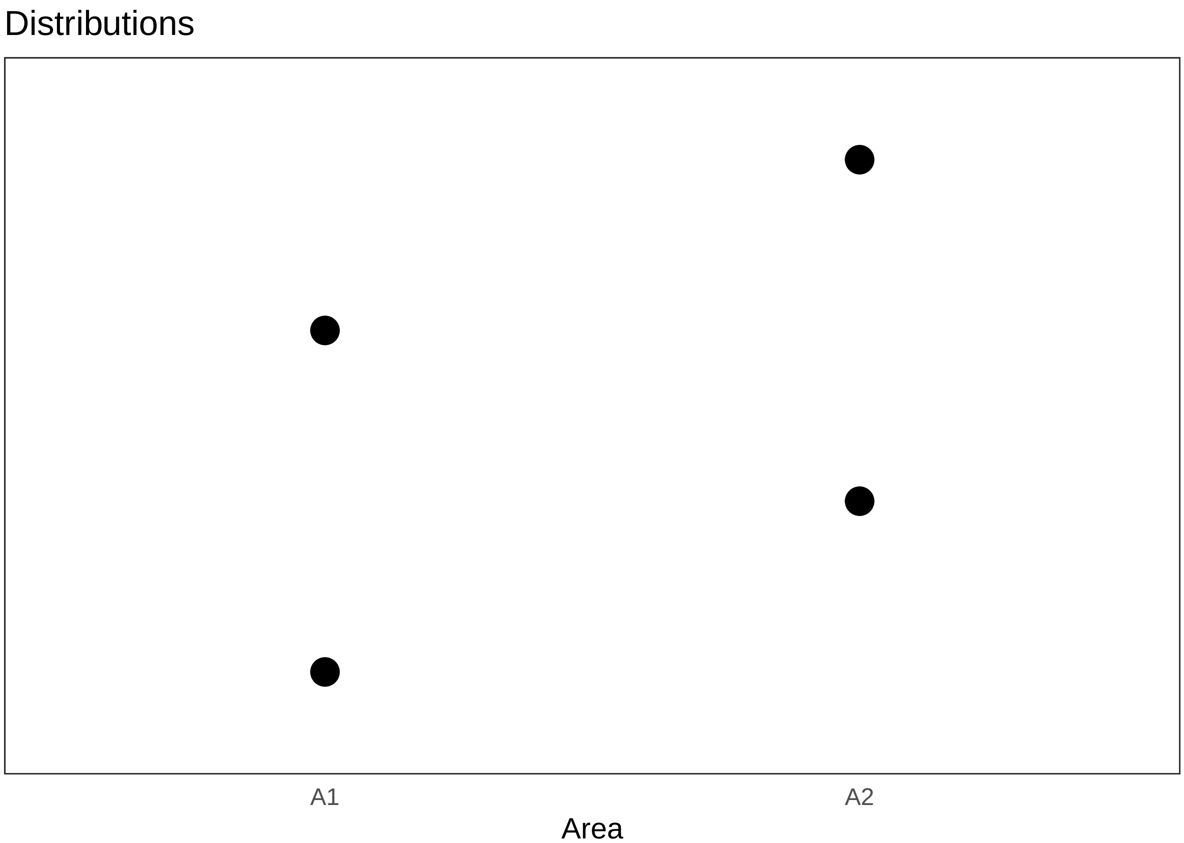
```
##   Terminal Area
## 1       t1   A1
## 2       t3   A1
## 3       t2   A2
## 4       t4   A2
```

```r
## plotting

## the tree, using ggtree

plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted")  +
             geom_tiplab(size=4.5, color="black")                +
             theme_tree2()                                       +
             ggtitle("Four terminals, all branches are equal to 1")



print(plotTree)
```

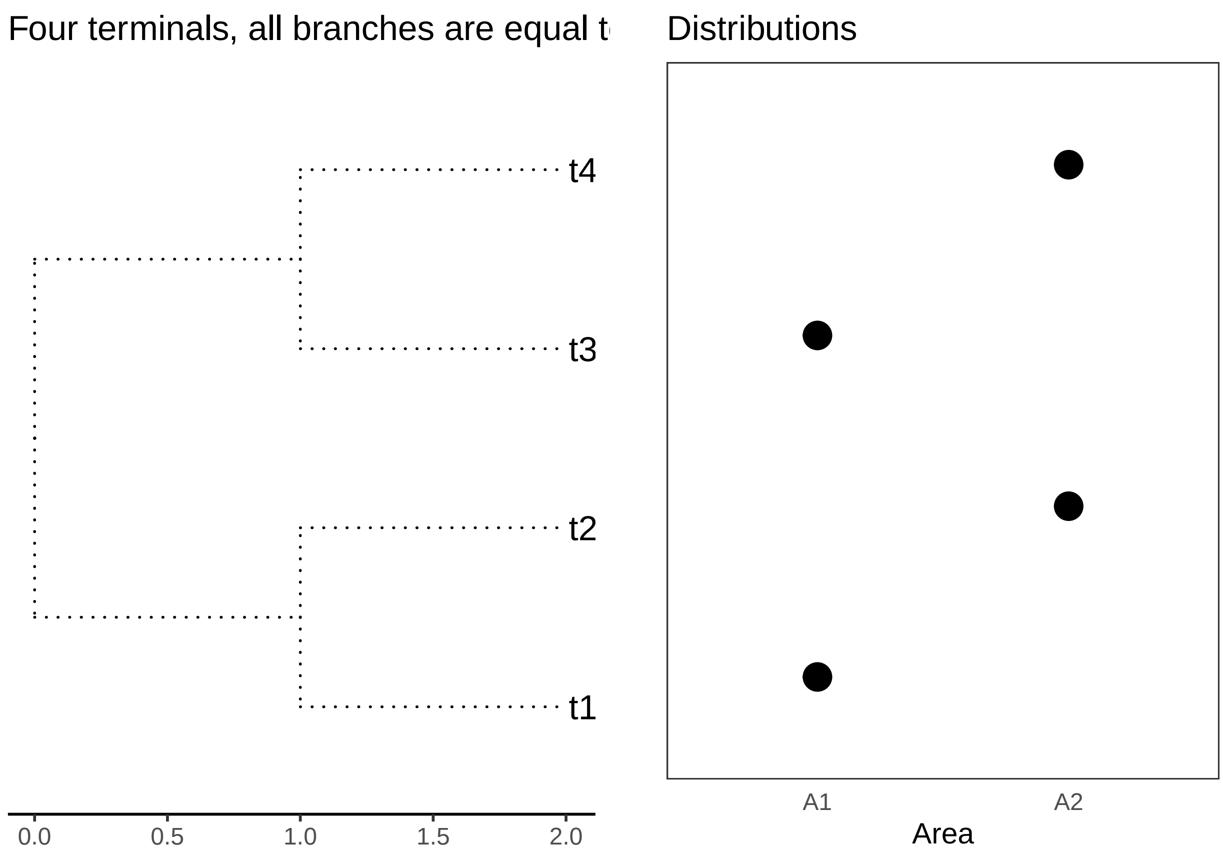
Four terminals, all branches are equal to 1




```
## The distribution using ggplot


plotDistrib <- ggplot(data=distXY,
                      aes(x= Area, y= Terminal , size =150))    +
               geom_point()                                      +
               theme_bw()                                        +
               theme( axis.text.y = element_blank(),
                      panel.grid.major = element_blank(),
                      panel.grid.minor = element_blank(),
                      axis.line = element_line(colour = NA_character_),
                      axis.ticks = element_blank(),
                      legend.position = "none" )               +
               labs(title = "Distributions",
                    y = "",
                    x = "Area")


##
print(plotDistrib)
```




```
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```




```
# Check whether names in both objects, initialTree and dist4taxa, are the same.


##all(colnames(dist4taxa) == initialTree$tip.label)

all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

```
#Report the branch length, and calculate the PD values.


initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)

initialPD
```

```
## [1] 4 4
```

```
#As expected, there is a tie between both areas.


#Now, we can see the impact of swapping the branch lengths; we will use the three
#available models to swap: "simpleswap", "allswap", "uniform", and we will swap the
#three classes of branches: "terminals", "internals", "all".
```

```r
##?swapBL


for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

 cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                         branch = rama
                        )
       printSwapBL(val, compact=TRUE)

        cat( "\n\n" )

     }

  }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
```

```
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1A2
## 1  100
```

```
#As expected, as all braches are EQUAL, nothing happens


#Now, let's see if the results could be related to the branch length (or not).



## we could use a single command (*evalBranch*)


##?evalBranch

AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")


## The structure of the output

##str(AllTerminals)

## Print it as a friendly output

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    1        A1A2        A2   lower     -1
## 2    2        A1A2        A1   lower     -1
## 3    3        A1A2        A2   lower     -1
## 4    4        A1A2        A1   lower     -1
## 5    1        A1A2        A1   upper      1
## 6    2        A1A2        A2   upper      1
## 7    3        A1A2        A1   upper      1
```

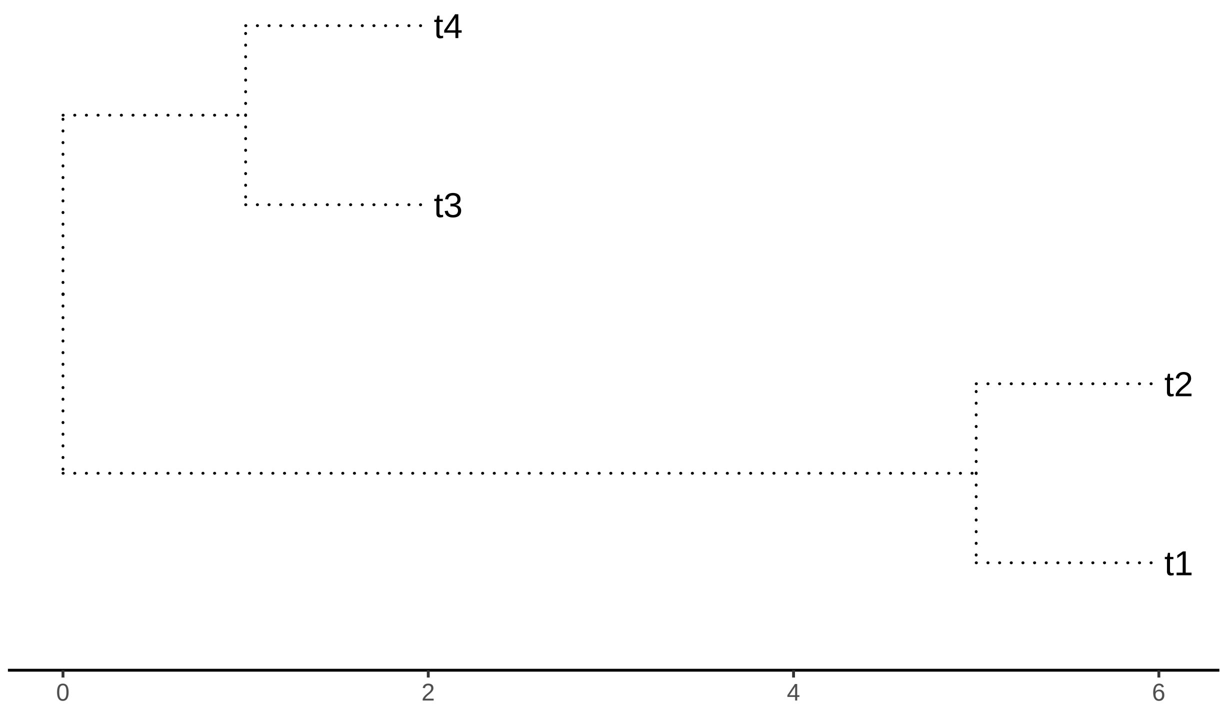
```
## 8    4         A1A2        A2    upper      1
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6        A1A2      A1A2   lower      0
## 2    7        A1A2      A1A2   lower      0
## 3    6        A1A2      A1A2   upper      0
## 4    7        A1A2      A1A2   upper      0
```

```r
#Now with a long branch


initialTree <- tree

initialTree$edge.length[1] <- 5

plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted")    +
             geom_tiplab(size=4.5, color="black")                  +
             theme_tree2()                                          +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=5. 2=1.




```
for( modelo in c("simpleswap","allswap","uniform") ){

      for( rama in c("terminals","internals","all") ){

         cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")

          val <- swapBL(tree=initialTree,
                        distribution = dist4taxa,
                        model = modelo,
                        branch = rama
                       )

      printSwapBL(val, compact=TRUE)

       cat( "\n\n" )

   }

 }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
```

```
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 10   73 17
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 41   32 27
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1  A2
## 1 43 57
```

```
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

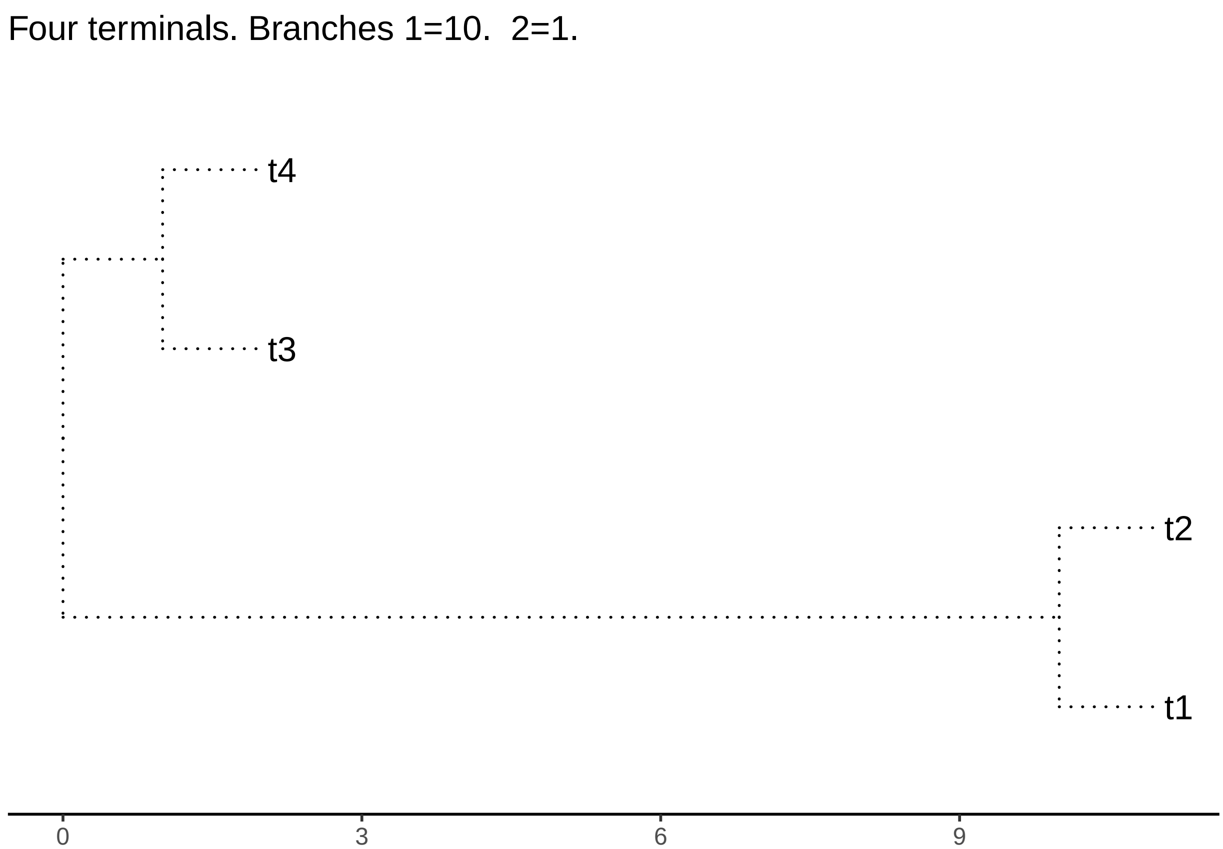
```
##   node initialArea FinalArea Aproach %Delta
```

```
## 1    1         A1A2        A2   lower     -1
## 2    2         A1A2        A1   lower     -1
## 3    3         A1A2        A2   lower     -1
## 4    4         A1A2        A1   lower     -1
## 5    1         A1A2        A1   upper      1
## 6    2         A1A2        A2   upper      1
## 7    3         A1A2        A1   upper      1
## 8    4         A1A2        A2   upper      1
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6         A1A2      A1A2   lower      0
## 2    7         A1A2      A1A2   lower      0
## 3    6         A1A2      A1A2   upper      0
## 4    7         A1A2      A1A2   upper      0
```

```r
# Now with a longer branch


initialTree <- tree

initialTree$edge.length[1] <- 10


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted")       +
             geom_tiplab(size=4.5, color="black")                      +
             theme_tree2()                                             +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```




```
for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                         branch = rama
                        )
       printSwapBL(val, compact=TRUE)

        cat( "\n\n" )

    }

 }
```

```
##
##    Branchs swapped=   terminals.  model to test simpleswap reps 100
##    A1A2
## 1  100
##
##
##
##    Branchs swapped=   internals.  model to test simpleswap reps 100
```

```
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 11   76 13
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 32   37 31
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 52 48
```

```r
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    1        A1A2        A2   lower     -1
## 2    2        A1A2        A1   lower     -1
```

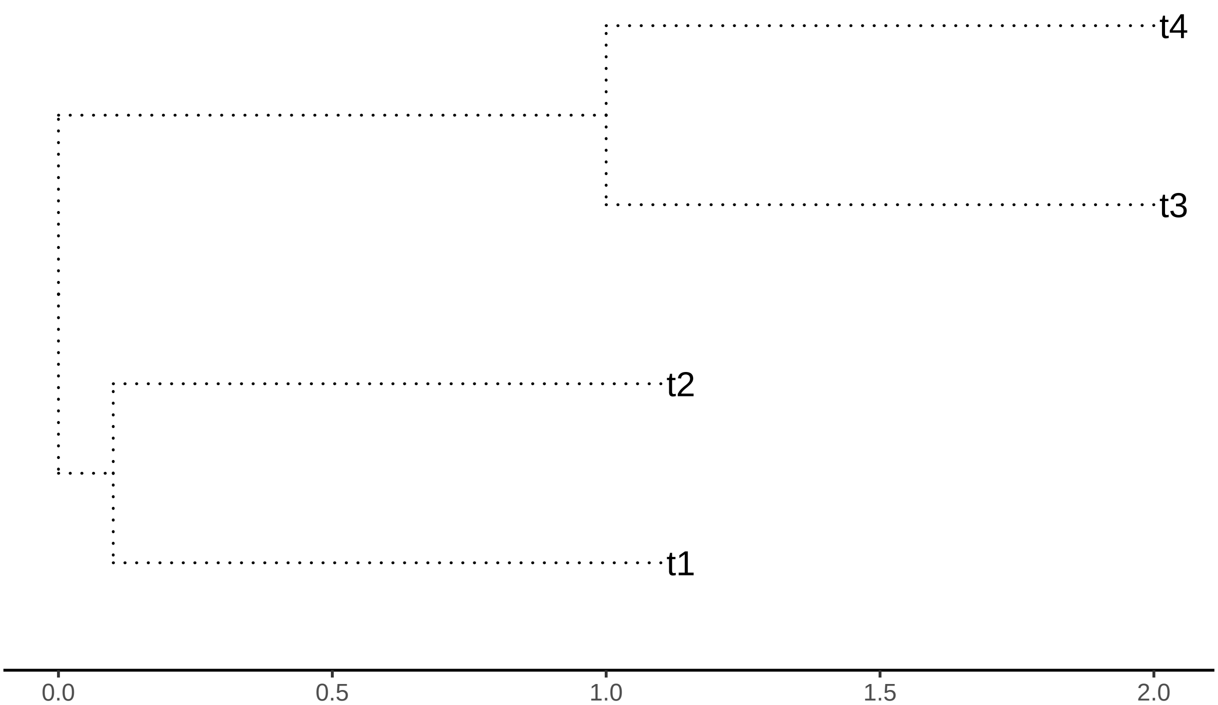
```
## 3    3         A1A2        A2   lower     -1
## 4    4         A1A2        A1   lower     -1
## 5    1         A1A2        A1   upper      1
## 6    2         A1A2        A2   upper      1
## 7    3         A1A2        A1   upper      1
## 8    4         A1A2        A2   upper      1
```

```
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6        A1A2      A1A2   lower      0
## 2    7        A1A2      A1A2   lower      0
## 3    6        A1A2      A1A2   upper      0
## 4    7        A1A2      A1A2   upper      0
```

```
# With a short branch



initialTree <- tree

initialTree$edge.length[1] <- 0.1


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted")  +
             geom_tiplab(size=4.5, color="black")                +
             theme_tree2()                                       +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=0.1. 2=1.




```
for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")

           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                         branch = rama
                        )

       printSwapBL(val, compact=TRUE)

        cat( "\n\n" )

    }

 }
```

```
##
##    Branchs swapped=   terminals.  model to test simpleswap reps 100
##    A1A2
## 1  100
##
##
```

```
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 15   71 14
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 35   28 37
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 49 51
```

```
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
```

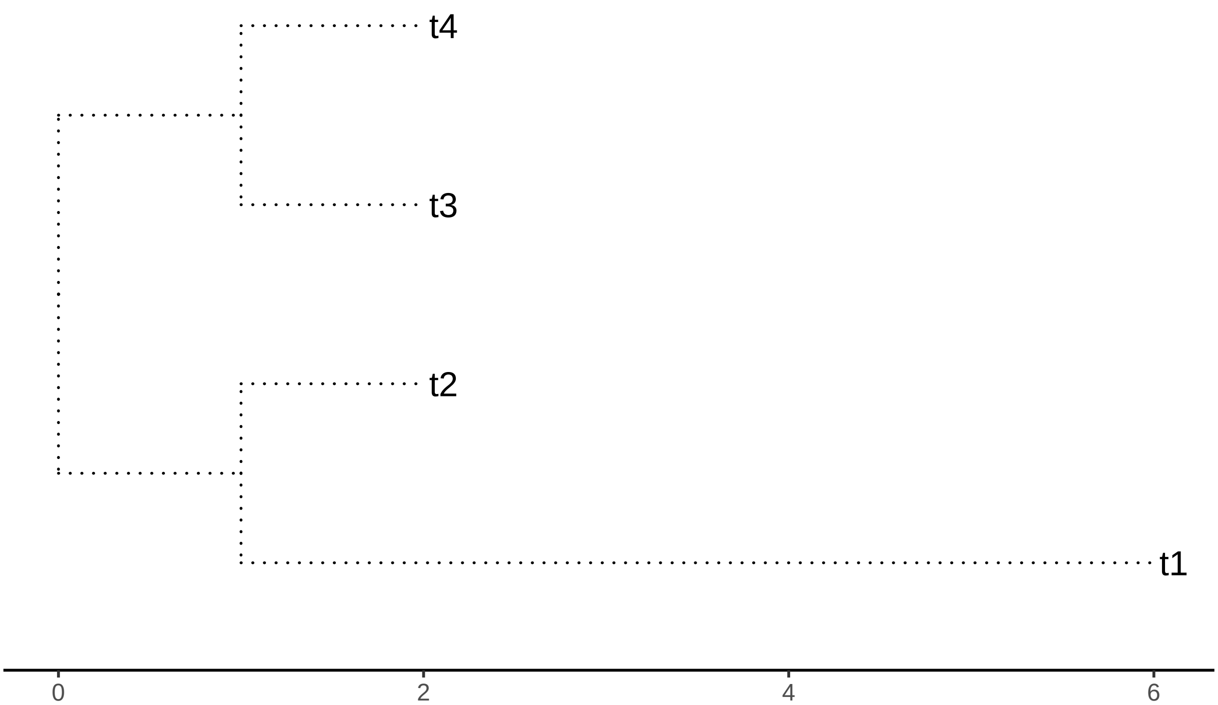
```
## 1    1         A1A2        A2   lower     -1
## 2    2         A1A2        A1   lower     -1
## 3    3         A1A2        A2   lower     -1
## 4    4         A1A2        A1   lower     -1
## 5    1         A1A2        A1   upper      1
## 6    2         A1A2        A2   upper      1
## 7    3         A1A2        A1   upper      1
## 8    4         A1A2        A2   upper      1
```

```
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6         A1A2      A1A2   lower      0
## 2    7         A1A2      A1A2   lower      0
## 3    6         A1A2      A1A2   upper      0
## 4    7         A1A2      A1A2   upper      0
```

```
#With a long branch in branch 2


initialTree <- tree

initialTree$edge.length[2] <- 5

plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted")  +
             geom_tiplab(size=4.5, color="black")                +
             theme_tree2()                                       +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=1. 2=5.




```r
for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")

           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                         branch = rama
                        )

       printSwapBL(val, compact=TRUE)

        cat( "\n\n" )

    }

 }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1 A2
## 1 66 34
##
##
```

```
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 64   15 21
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1 A2
## 1 55 45
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 27   39 34
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1 A2
## 1 52 48
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 59 41
```

```
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
```

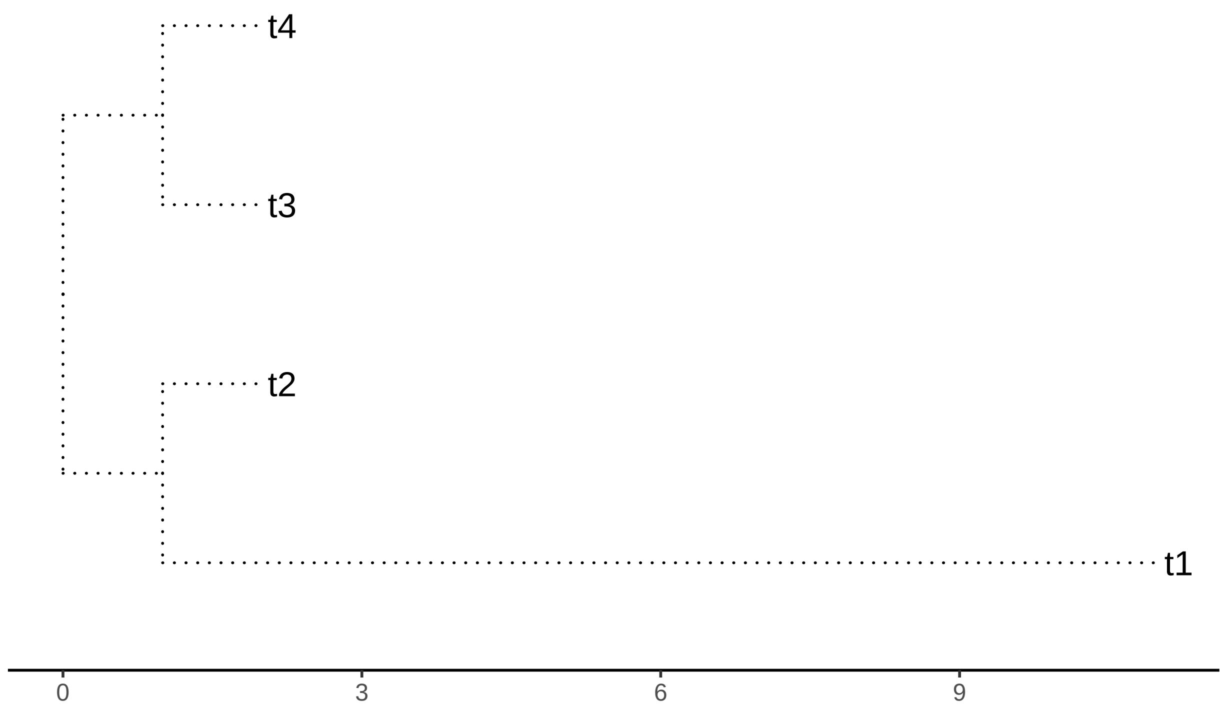
```
## 1    1          A1        A2   lower    -80
## 2    2          A1        A1   lower      0
## 3    3          A1        A1   lower      0
## 4    4          A1        A1   lower      0
## 5    1          A1        A1   upper      0
## 6    2          A1        A2   upper    400
## 7    3          A1        A1   upper      0
## 8    4          A1        A2   upper    400
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6          A1        A1   lower      0
## 2    7          A1        A1   lower      0
## 3    6          A1        A1   upper      0
## 4    7          A1        A1   upper      0
```

```r
# Now with a longer branch2


initialTree <- tree

initialTree$edge.length[2] <- 10


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted")  +
             geom_tiplab(size=4.5, color="black")                +
             theme_tree2()                                       +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=1. 2=10.




```
for( modelo in c("simpleswap","allswap","uniform") ){

      for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")

          val <- swapBL(tree=initialTree,
                        distribution = dist4taxa,
                        model = modelo,
                        branch = rama
                       )

      printSwapBL(val, compact=TRUE)

       cat( "\n\n" )

   }

 }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1 A2
## 1 68 32
##
##
```

```
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 76    8 16
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1 A2
## 1 48 52
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 35   31 34
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1 A2
## 1 54 46
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 46 54
```

```
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

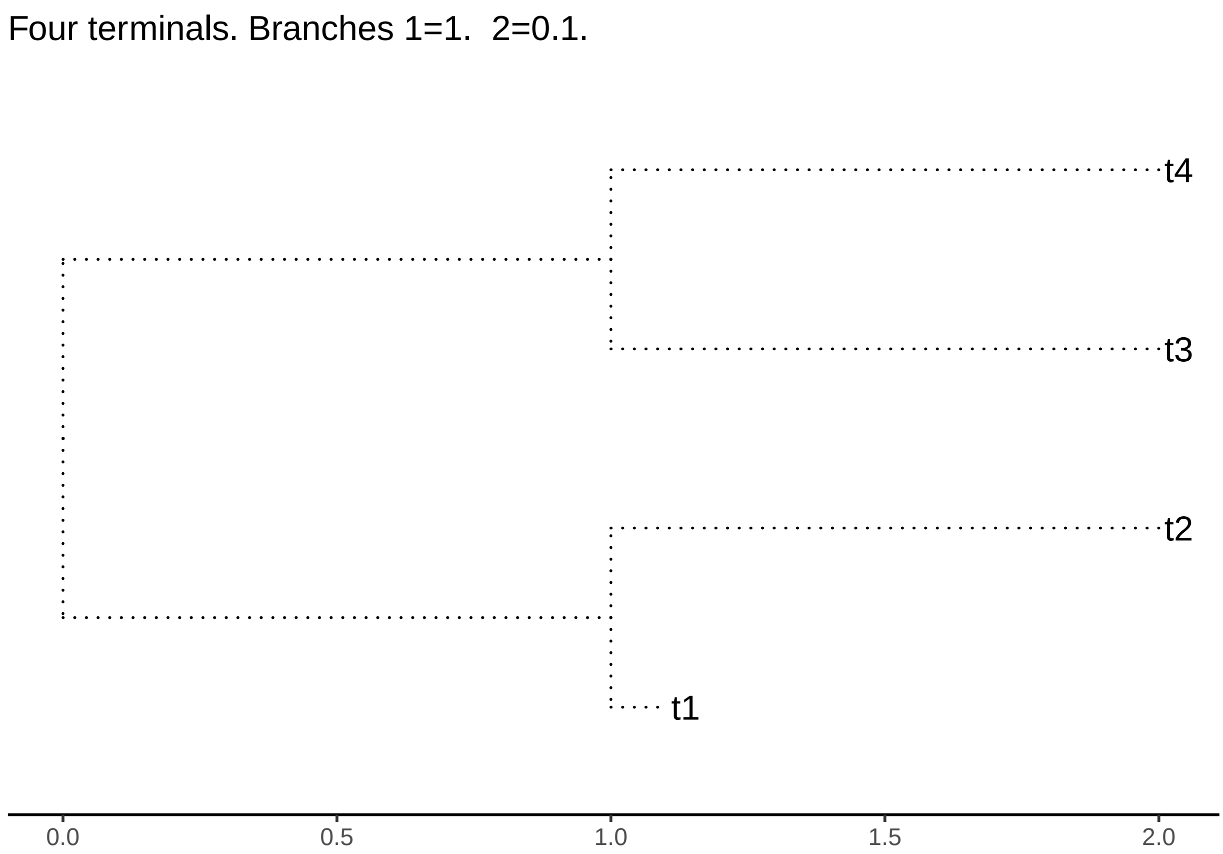
```
##   node initialArea FinalArea Aproach %Delta
```

```
## 1    1          A1        A2   lower    -90
## 2    2          A1        A1   lower      0
## 3    3          A1        A1   lower      0
## 4    4          A1        A1   lower      0
## 5    1          A1        A1   upper      0
## 6    2          A1        A2   upper    900
## 7    3          A1        A1   upper      0
## 8    4          A1        A2   upper    900
```

```
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6          A1        A1   lower      0
## 2    7          A1        A1   lower      0
## 3    6          A1        A1   upper      0
## 4    7          A1        A1   upper      0
```

```
#now with a shorter branch



initialTree <- tree

initialTree$edge.length[2] <- 0.1


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted")              +
             geom_tiplab(size=4.5, color="black")                            +
             theme_tree2()                                                   +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```




```
for( modelo in c("simpleswap","allswap","uniform") ){

      for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")

          val <- swapBL(tree=initialTree,
                        distribution = dist4taxa,
                        model = modelo,
                        branch = rama
                       )

      printSwapBL(val, compact=TRUE)

       cat( "\n\n" )

   }

 }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1 A2
## 1 32 68
##
##
```

```
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
##    A2
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 17   14 69
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1 A2
## 1 52 48
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##    A2
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 31   40 29
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1 A2
## 1 51 49
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##    A2
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 49 51
```

```r
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
```

```
## 1    1           A2        A2   lower      0
## 2    2           A2        A1   lower    -90
## 3    3           A2        A2   lower      0
## 4    4           A2        A1   lower    -90
## 5    1           A2        A1   upper    900
## 6    2           A2        A2   upper      0
## 7    3           A2        A1   upper     90
## 8    4           A2        A2   upper      0
```

```
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6           A2        A2   lower      0
## 2    7           A2        A2   lower      0
## 3    6           A2        A2   upper      0
## 4    7           A2        A2   upper      0
```